A complete ten-variable consistency equation

Strictly positive integer unknowns; exact degree 328; an 85-operation evaluator.

The only unknowns are

$$J, F, \alpha, i, T, w, Y, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

All other symbols in the equation are polynomial abbreviations:

$$\begin{gathered}
q_- = \beta J, \quad q = q_- + 1, \\
C = q - F - \alpha, \\
X = wq, \quad k = \eta + \zeta, \quad c = kY + \eta, \\
\mu = q^2 + 1, \quad A = \mu Y(X+1), \quad H = 4A - 1, \\
\Delta = A(A+2), \quad U = ic^2, \\
V_0 = X(\mu Y)^2, \quad \ell = V_0 k, \quad E = \ell(\ell + k) + 1, \\
R = [q_-(q-F) + \alpha](q^2 - 1) + (M_C + qM_F)J, \\
e = k - R - 1, \quad B_t = (K + w)C - F, \\
a = \Delta U^2 E, \quad D = 1 + a, \quad G = ED, \\
S = 1 + 4aD, \quad Q = 4\Delta aD, \\
b = c(T + \alpha) - R, \quad V = Sb - c, \\
K_2 = X(3c - 2X) + 2, \\
P_1 = q^3 q_-, \quad M = P_1 \mu^2 H, \quad R_G = M^2 G, \\
Z = P_1 \mu^2 [H(n - bVB_t) + K_2] \\
\quad + H[\mu^2(q_- Y - q^3 B_t) - P_1 e].
\end{gathered}$$

The single equation is

$$\boxed{Q(Z^2 - R_G V^2) + (R_G - Z^2) = 0.}$$

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum\nolimits_{j=0}^{m-1} x^j.$$

Define

$$\begin{aligned}
\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1011\} \\
& \cup \{1013, \dots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.
\end{aligned}$$

Then

$$\begin{aligned}
M_C &= B - 1 - \sum\nolimits_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\
M_F &= B - 1 + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) \\
&\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\
&\quad + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805}, \\
K &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\
&\quad + (\rho^{4393} + \rho^{5495})\Phi(1/\rho),
\end{aligned}$$

where

$$\begin{aligned}
\Phi(x) &= (1 + 2^{51}) G_{943}(x) - 2^{10} \sum\nolimits_{i=0}^{942} v(t_i) x^i \\
&\quad + \sum\nolimits_{(b,e,m) \in \mathcal{C}} 2^b x^e G_m(2x), \\
\mathcal{C} &= \{(10, 1023, 8), (18, 1013, 9), (27, 1003, 8), \\
&\quad (35, 983, 9), (44, 1031, 1), (45, 1022, 1), (46, 1011, 1), \\
&\quad (47, 992, 1), (48, 1095, 1), (49, 2195, 1), (50, 3296, 1)\}.
\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. Let

$$\begin{gathered}
h(q,b) = 2(q+1) + b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\}, \\
\mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\}, \\
r_q = 6 + |\mathcal{R} \cap \{0, \dots, q-1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0, \dots, q-1\}|.
\end{gathered}$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w, n, e, s)$
$0, 1$	$(2, 435, 3, 1),\ (5, 0, 0, 0)$
$j + 2\ (j = 2q + b < 430)$	$((1 - d_j) l_{u_j}, h(q,b), d_j r_{u_j}, v_j)$
$432 + b\ (b = 0, 1)$	$(0, h(215, b), 0, h(215, b))$
$434 + b\ (b = 0, 1)$	$(0, b, 0, b)$
$424 + 2r_q + b\ (q \in \mathcal{R}, b = 0, 1)$	$(r_q, b, 0, h(q, b))$
$424 + 2l_q + b\ (q \in \mathcal{L}, b = 0, 1)$	$(0, b, l_q, h(q, b))$
$924 + j\ (0 \le j < 4)$	$(0, j + 430\lfloor j/2 \rfloor, 0, 438)$
928	$(2, 434, 2, 0)$
929	$(3, 436, 4, 4)$
930	$(4, 436, 4, 0)$
$497 + t\ (t = 434, 435, 436)$	$(1, 438, 1, t)$
$^{934+3a+b-6\lfloor a/4 \rfloor}_{a \in \{0,1,4\},\ b=0,1,2}$	$(a, 437, b, 437)$

For $t = (w, n, e, s)$ put

$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\, 2^{34}.$$

A closed decoder for the 430 actions. Put $D_b(A, k) = \lfloor A / b^k \rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$\begin{gathered}
N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \lfloor \mathcal{A} / 2^{636} \rfloor \bmod 2^{1675}, \\
E_0 = \lfloor \mathcal{A} / 2^{2311} \rfloor, \\
\nu_j = D_2(N, j), \quad \sigma_j = \sum\nolimits_{k=0}^{j-1} \nu_k, \\
m_q = D_2(N, 430 + q), \quad \tau_q = \sum\nolimits_{k=0}^{q-1} m_k, \\
F_q = 14(1 - m_q) + m_q D_{16}(E_0, \tau_q), \\
u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j) D_{216}(T_0, j - \sigma_j), \\
v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).
\end{gathered}$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \dots, 61$, is `0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz`.

Concatenate the following 436 characters as digits $e_0, \dots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

The ordinary state/color-conjugacy, native and Pell arguments give unsolvability iff Con(ZFC). Native calibration and internal-ZFC formalization remain open in Lean. This page is a complete finite specification, not an unsolvability proof.